FOSSIL FUELS & INDUSTRY · 1750–2024

Global CO_2 Emissions: Who Emits, How Much, and What's Changing

A data-forward overview of emissions by country, region, and per capita — from historical growth to 2024 shifts.

Scope

CO_2 from fossil fuels and industry; land-use change noted separately.

Audience

Policy analysts, climate negotiators, and sustainability researchers.

Source

Global Carbon Budget 2025 via Our World in Data.

LONG-RUN TRAJECTORY

Global CO_2 Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

Global fossil-fuel & industry CO_2 emissions, approximate billion tonnes per year

Source: Global Carbon Budget 2025 via Our World in Data; values shown are those specified in source brief.

35+ Bt

Annual global fossil-fuel and industry CO_2 emissions today.

~6 Bt

World emissions in 1950.

>20 Bt

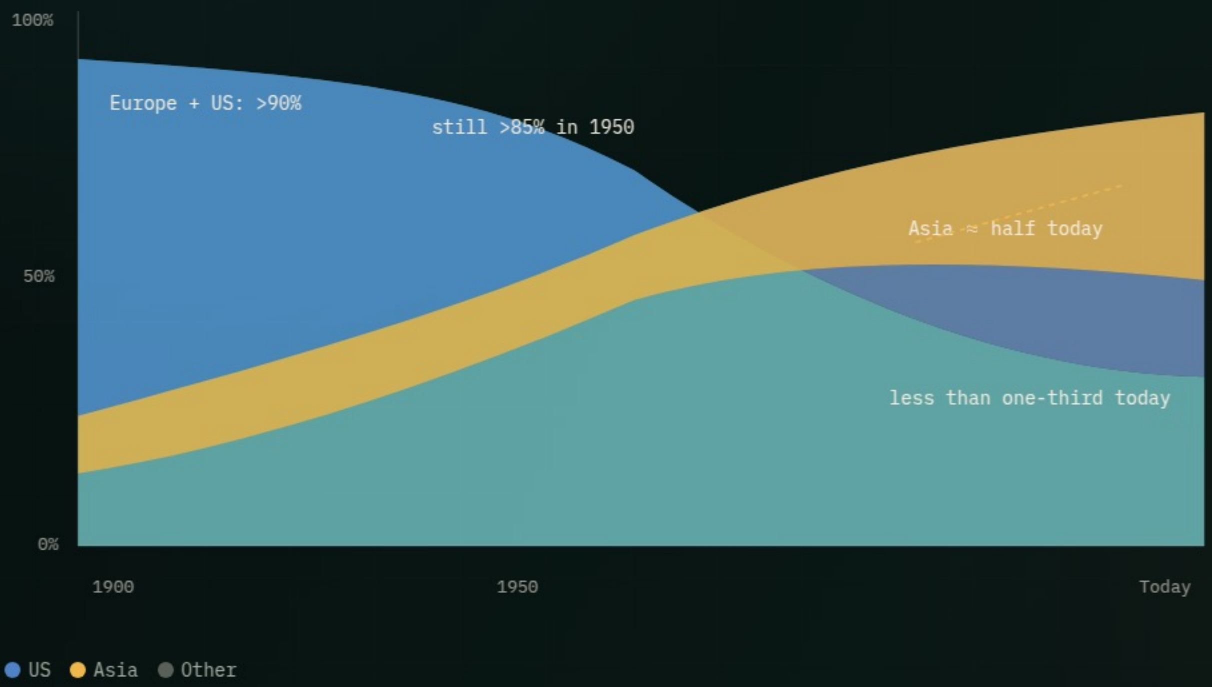
By 1990, nearly quadrupled.

Interpretation: growth has slowed in recent years, but the global curve still has not peaked.

REGIONAL SHARES

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

Conceptual stacked-area view, using specified share thresholds



>90%
Europe and the United States in 1900.

>85%
Europe and the United States in 1950.

< one-third
Europe and the United States today.

≈ half
Asia's current share, led by China.

COUNTRY-LEVEL DOMINANCE

China Now Emits More Than One-Quarter of Global CO_2

Treemap of approximate 2024 shares of global CO_2 emissions

China
~31%
Rising, growth slowing

US
~13%
Declining

India
~8%
Rising

EU-27
~6%
Declining

Russia
~5%
Roughly stable

Other shares
Africa total ~3–4%; South America total ~3–4%; international aviation & shipping ~3%.

Areas are visual emphasis, not a full global accounting. Values come directly from the source brief.

~31%
China's approximate share of global CO_2 emissions in 2024.

Africa and South America each account for only **~3–4%** — comparable to international aviation and shipping combined at **~3%**.

PER CAPITA INEQUALITY

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

Approximate 2024 tonnes CO_2 per person; selected countries from source brief

Scale extends to ~15 t/person to reflect the source brief's ~14–15 t high-emitter range.

150×

Approximate gap between lowest-emitting nations and high-emitting large countries.

The average American or Australian emits in under two days what the average person in Mali or Niger emits in an entire year.

France and the UK are much lower than the US despite similar living standards, according to the source brief.

CURRENT PER-PERSON COMPARISON

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

Snapshot comparison, approx. 2024 tonnes CO_2 per person

Reframing responsibility: China is the largest total emitter, but its per capita emissions remain below the US.

India remains very low at **~2 t/person**, far below the world average of **~4.5 t/person**.

Source brief references figure_8.jpg for additional historical context on per capita evolution.

2024 ANNUAL CHANGE

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

Diverging bar / choropleth placeholder based on source brief descriptions

United States declines	blue shading ↓ emissions	
Much of EU declines	2024 annual % change	**Russia increases**
United Kingdom declines	orange/red shading ↑ emissions	**Parts of Southeast Asia increase**
	large increases >10%	**Turkmenistan / Uzbekistan region and several African nations**

No country-specific percentages are added beyond the source brief; “>10%” appears only for the Central Asia region described there.

Mixed signals

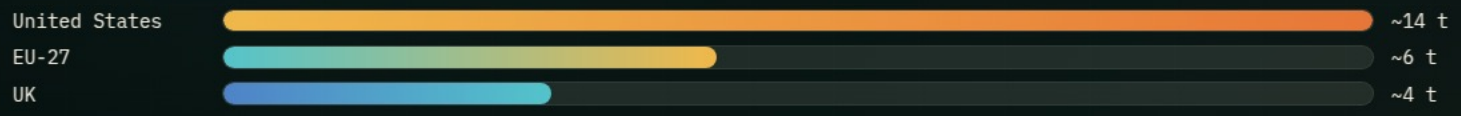
Western declines coexist with increases in parts of Asia, the Middle East, and Africa.

The 2024 pattern shows why global totals can keep rising even when many developed nations reduce emissions.

ENERGY MIX MATTERS

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

Comparative bar chart plus qualitative energy-mix explanation



Lower per capita

France, the UK, and Portugal: higher share of electricity from nuclear and renewable sources.

Higher per capita

Germany, the Netherlands, or the US: more fossil-fuel dependence in the energy mix.

Among wealthy nations with similar prosperity levels, per capita emissions vary significantly because of energy mix choices.

Policy lever

Technology and electricity choices can decouple prosperity from emissions.

This slide intentionally avoids a detailed energy breakdown because the source brief provides qualitative energy-mix insight, not fuel-share data.

SYNTHESIS

Key Takeaways: Responsibility Is Shared but Unequal

01 Global fossil-fuel CO_2 emissions exceed 35 Bt/yr and have not yet peaked.

02 China is the largest annual emitter at roughly 31%, but per capita emissions remain below the US.

03 Historical responsibility is dominated by Europe and the US: >90% in 1900, but less than one-third today.

04 Per capita gaps of roughly 150× persist between the lowest-emitting and highest large-country emitters.

05 Energy policy and technology choices can decouple prosperity from emissions.